

5(a)	1 four DNA bases named correctly ; <i>correct spellings</i> 2 purine or pyrimidine column correct for A, C, G, T ; 3 final column correct ; 4 uracil (<i>spelled correctly</i>) <u>and</u> pyrimidine ; if 0 or 1 mark gained, can increase to max 2: • if 0 marks, can check for two correct rows • only mp 1, 2 or 3 correct, check for one correct row • only mp 4 correct, check for a correct A / C / G / T row <table><tr><th>base</th><th>name of base</th><th>purine or pyrimidine</th><th>RNA or DNA or both</th></tr><tr><td>A</td><td>adenine</td><td>purine</td><td>both</td></tr><tr><td>C</td><td>cytosine</td><td>pyrimidine</td><td>both</td></tr><tr><td>G</td><td>guanine</td><td>purine</td><td>both</td></tr><tr><td>T</td><td>thymine</td><td>pyrimidine</td><td>DNA</td></tr><tr><td>U</td><td>uracil</td><td>pyrimidine</td><td>RNA</td></tr></table>	base	name of base	purine or pyrimidine	RNA or DNA or both	A	adenine	purine	both	C	cytosine	pyrimidine	both	G	guanine	purine	both	T	thymine	pyrimidine	DNA	U	uracil	pyrimidine	RNA	4
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5(b)(i)	purine because it is a, double ring structure / has two rings ; A purine because similar structure to, adenine / guanine	1																								

Question	Answer	Marks
5(b)(ii)	<i>mp1 / 2 / 4 idea that carbovir triphosphate is more similar to the (activated) DNA nucleotides that are added to the elongating chain by DNA polymerase</i> <i>mp3 justifying why it is better to have a nucleotide analogue (carbovir triphosphate) by explaining the role of DNA nucleotides</i> <i>look for ora for abacavir</i> <i>any two from:</i> 1 carbovir triphosphate is (similar to / same as), an activated nucleotide / a nucleotide / not a nucleoside ; A similar to, A / G, nucleotides I ref. to purine 2 carbovir triphosphate, has (three) phosphates / is activated / is phosphorylated ; 3 detail of, (DNA) polymerase action / (activated DNA) nucleotides ; e.g. (nucleotides are) added to the, growing / elongating, chain (nucleotides) form H bonds with, complementary base / AW (nucleotides are) substrates for DNA polymerase phosphodiester bonds form between nucleotides <i>allow mp3 in context of carbovir triphosphate (link to mp1)</i> 4 <i>ref. to triangle / triangular cycloalkane, no longer present / not on DNA nucleotides ;</i> 5 AVP ; e.g. carbovir triphosphate better fit to active site of enzyme / AW	2

Question	Answer	Marks
5(b)(iii)	any four from: <i>carbovir triphosphate can be inserted into a growing chain...</i> - prevents polymerase from adding DNA nucleotide to growing chain / AW ; A <i>idea of</i> complete (viral) DNA not made if, chain does not elongate / no more nucleotides added R adding base R <i>if in context of stop codon / transcription</i> I <i>ref. to</i> complementary base pairing - similar <u>shape</u> to, substrate / (activated / phosphorylated) nucleotide ; A complementary shape to active site of, DNA polymerase / enzyme - acts as an inhibitor ; 4,5 further detail of how, inhibitor / carbovir triphosphate, may act ;; <i>two from = 2 marks</i> e.g. may be irreversible fits into / binds to, active site of, enzyme / DNA polymerase competes with (DNA) nucleotide for active site R base fewer ES complexes form <i>in context of enzyme with DNA nucleotide</i> <i>idea of</i> fewer enzyme molecules available to bind to nucleotides may cause change in, shape / tertiary structure / 3D structure, of (DNA) polymerase / enzyme I <i>ref. to</i> complementary base pairing (DNA polymerase) may not form phosphodiester bonds ; R between, bases / complementary bases / complementary nucleotides <i>ref. to</i> proofreading mechanism ; e.g. error may cause breakdown of (newly synthesised) DNA may be too similar to be noticed as error and not repaired - AVP ; e.g. (viral) DNA containing analogue may not be further replicated DNA polymerase cannot move along DNA template strand AW DNA polymerase may become detached from template strand <i>idea that</i> carbovir triphosphate, introduces a mutation / changes DNA formed / changes nucleotide sequence	4